AUTHORSHIP VERIFICATION · COMPUTATIONAL STYLOMETRY

Sen2Pro

A Probabilistic Perspective on Sentence Embedding for Authorship Verification

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0.989
AUC on the Shakespeare authorship question

90.3%
Balanced accuracy under leave-one-document-out

0.94 / 0.87
Shakespeare recall / impostor rejection

205K
50-word segments · 50 Shakespeare + 30 impostor authors

The Solution — How Sen2Pro Works

A reproducible batch pipeline integrating probabilistic sentence embeddings with the Deep-Impostors stylometric method.

01 **Segment text** — Documents cleaned of Gutenberg boilerplate, split into 50-word units. →

02 **Probabilistic embedding** — Each segment a Gaussian $N(\mu, \sigma^2)$ via all-distilroberta-v1 + MC-Dropout. →

03 **Impostor separators** — A CNN + BiLSTM separator trained per pair of impostor authors. →

04 **Stylistic signal** — Sliding windows turn probabilities into a smooth per-document signal. →

05 **Compare & cluster** — Normalized DTW aligns signals; K-medoids + Isolation Forest split the corpus. →

06 **Calibrated verdict** — Conformal predictor returns Authentic · Suspicious · Abstain · Outlier.

01 · BACKGROUND

The Shakespeare Authorship Question

Authorship verification asks whether a questioned document was written by a specific author from **style**, not content — central to forensic linguistics, literary scholarship and digital security.

Style is a subtle mix of features shaped by genre, period, collaboration and imitation, and drifts within one author over time. The Shakespeare canon pairs a rich corpus with a genuinely open question — an ideal stress test. Sen2Pro integrates two Phase-A strands: **Gaussian sentence embeddings** (Shen et al., 2023) and the **Deep-Impostors** framework (Volkovich & Avros, 2025).

DATA · CORPUS

What We Analysed

- **50 Shakespeare texts** + 30 impostor authors (110 texts) from Project Gutenberg.
- Contemporaries (Marlowe, Jonson, Milton), later canon (Austen, Dickens, Woolf) and moderns (Christie, Asimov, Clarke).
- **22,703** Shakespeare + **182,476** impostor 50-word segments.

02 · REQUIREMENTS

Requirements — All Met

- ✓ **FR1–2** Load, clean & segment raw text into 50-word units.
- ✓ **FR3** Probabilistic embeddings $N(\mu, \mathrm{diag}\,\sigma^2)$.
- ✓ **FR4** Segment classification (CNN + BiLSTM).
- ✓ **FR5–6** Temporal smoothing + anomaly detection.
- ✓ **NFR1–3** ~1 min / 2000-word doc · robust input · handles >512-token texts.

05–06 · CHALLENGES & INSIGHTS

- **Fixed 0.5 threshold** mis-rejected Shakespeare → calibrated to ≈0.91.
- **Log-variance diluted signal** → weighted + standardised; more features can hurt.
- **Honest evaluation beats a flattering number** — one LOO protocol + random control from day one.

03 · METHOD

Probabilistic Embeddings

Each segment is a Gaussian: the mean captures central style, the per-dimension variance captures uncertainty, estimated by the method of moments.

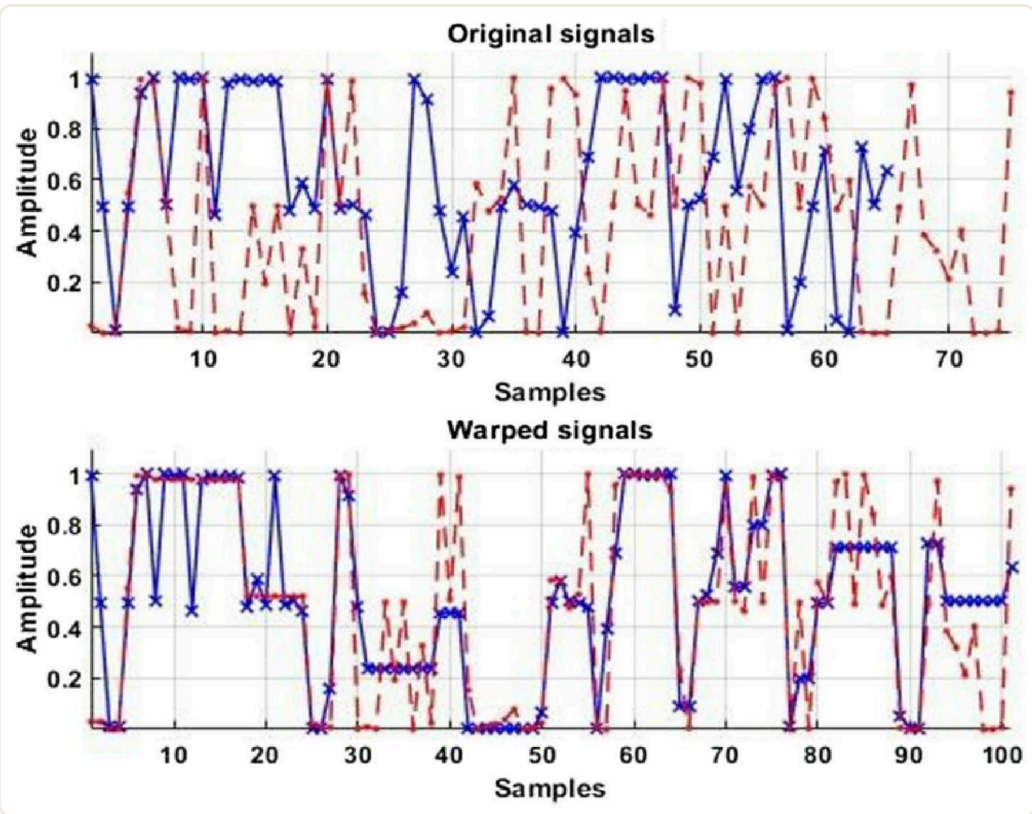
$$\mu = (1/N)\ \Sigma\ x_i \quad \cdot \quad \sigma^2 = (1/N)\ \Sigma\ (x_i - \mu)^2$$

- **Model uncertainty** — 25× Monte-Carlo Dropout passes.
- **Data uncertainty** — optional perturbations (substitution, deletion, swap, masking).

SEPARATION & VERIFICATION

From Signals to a Verdict

- **Separator** Conv1D → BiLSTM(500) → BiLSTM(250) → Dense, per impostor pair.
- **Stylistic signal** overlapping sliding windows (w=8, stride 1).
- **Normalized DTW** z-scored, path-normalized alignment feeds K-medoids.
- **Conformal predictor** gives coverage-guaranteed sets with abstain / outlier for disputed texts.



DTW warps two stylistic signals onto a common timeline before measuring distance.

04 · ENVIRONMENT

Tools & Environment

Python notebook on Google Colab Pro; a single hyperparameter cell is the source of truth, with μ/σ^2 cached to Drive for fast re-runs on an NVIDIA RTX PRO 6000 (102 GB, bf16).

`PyTorch 2.x · CUDA` `sentence-transformers` `HuggingFace` `scikit-learn` `fastdtw` `nltk` `numpy · scipy`

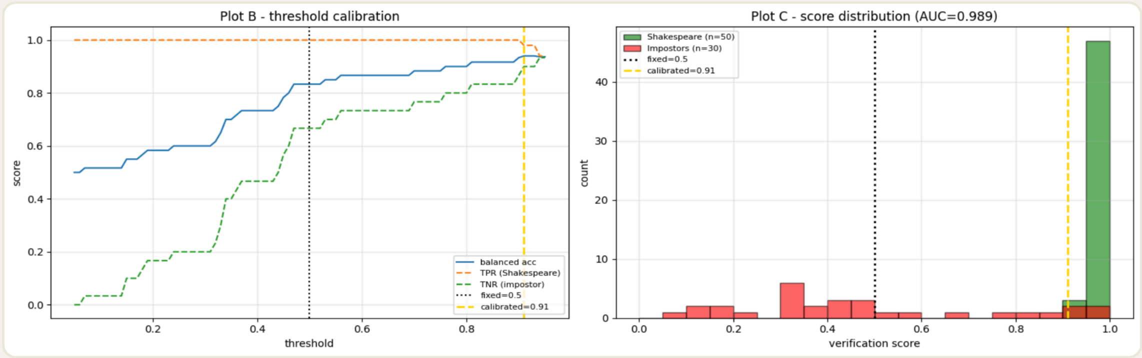


Fig 1 — Calibration. A fixed 0.5 cut is mis-calibrated for a rank score; the calibrated threshold ≈0.91 recovers Shakespeare recall at **AUC = 0.989**.

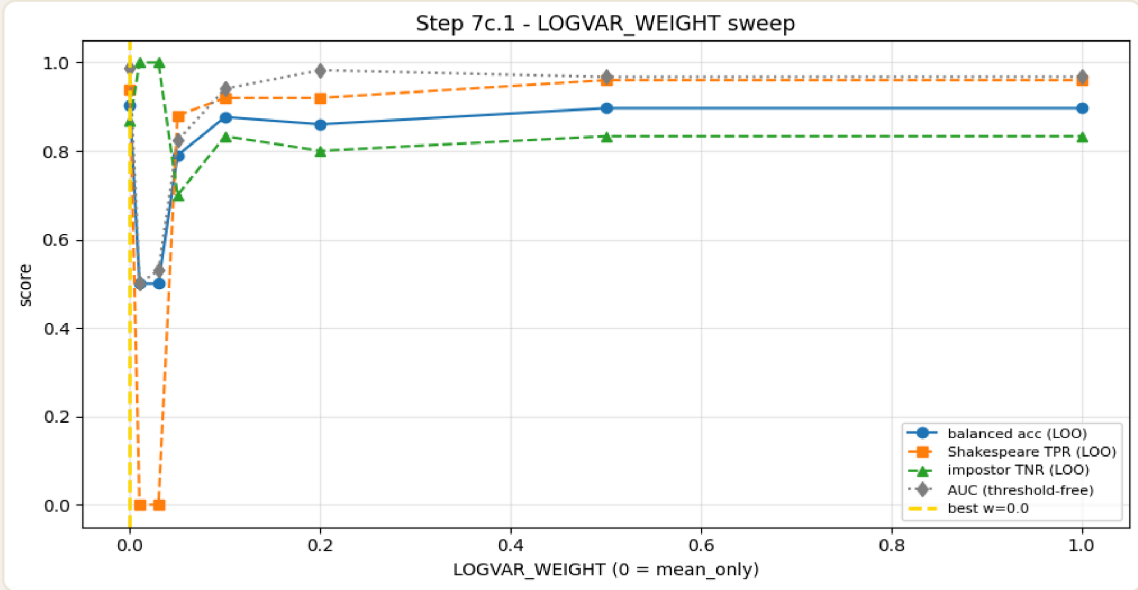


Fig 2 — Log-variance sweep. Balanced accuracy, recall, rejection and AUC all peak near **w = 0 (mean-only)**.

The honest finding

The deterministic mean embedding is already a very strong baseline. Encoder-level uncertainty adds **limited but non-zero** signal — mean-only and mean+logvar **tie at 90.3%** balanced accuracy. A defensible result, reported without over-claiming.

METRICS · TARGETS VS ACHIEVED

METRIC	ACHIEVED	STATUS
Accuracy >85%	AUC 0.989 · 90.3%	Met
Perturbation <10%	Stable (sweep)	Met
Signal smoothness	Sliding windows	Met
FPR <5%	0.133 at bal. point	Trade-off
Latency <200 ms	Met on GPU (bf16)	Met

Full Phase A & B books, complete code, poster and demo video are in the project repository. **Generative-AI tools** were used only for auxiliary support (summarisation, boilerplate, prose) — all algorithmic logic and verification by the team.

github.com/KfirCohen-PyLab/Final_Degree_Project

SCAN FOR CODE